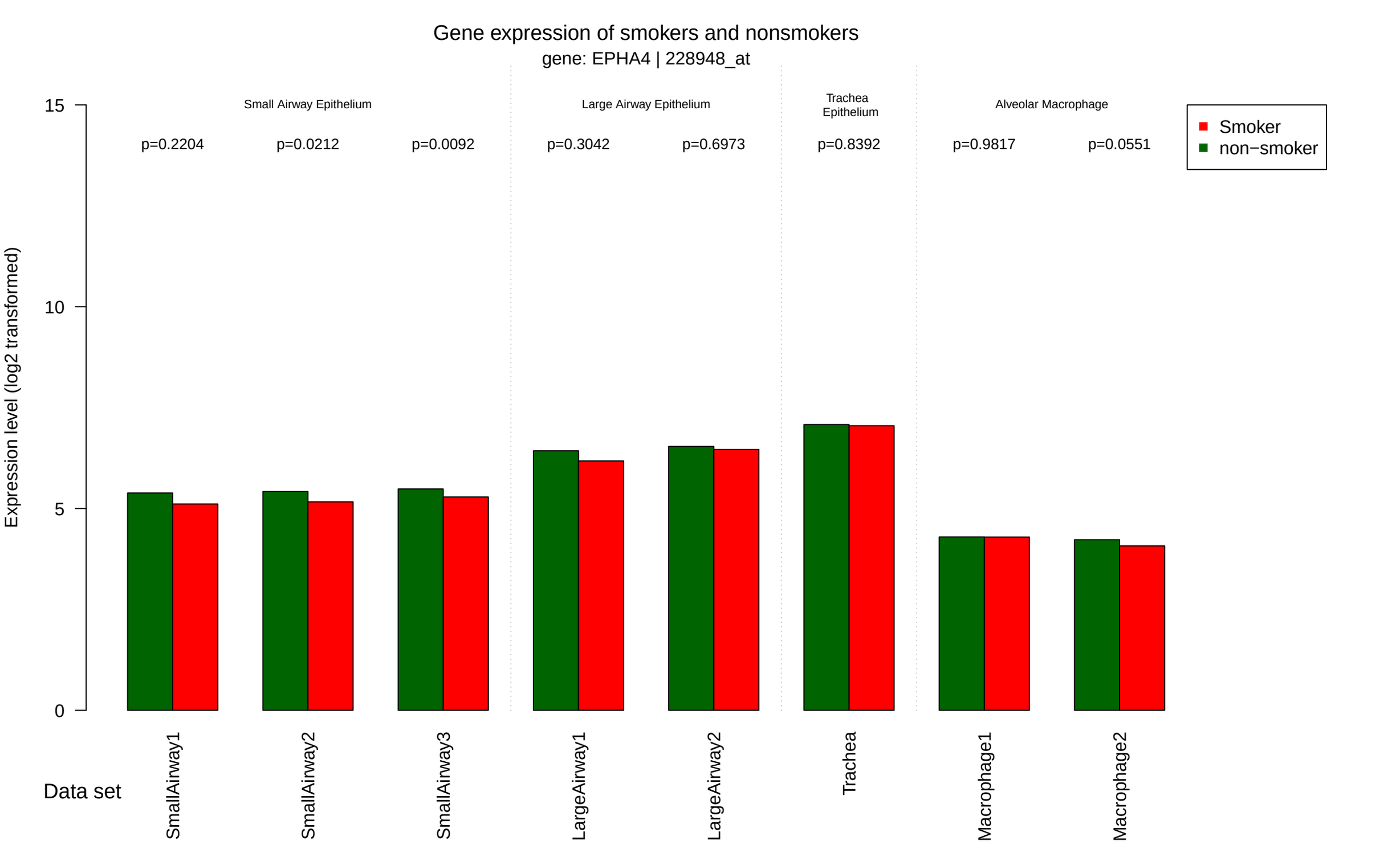
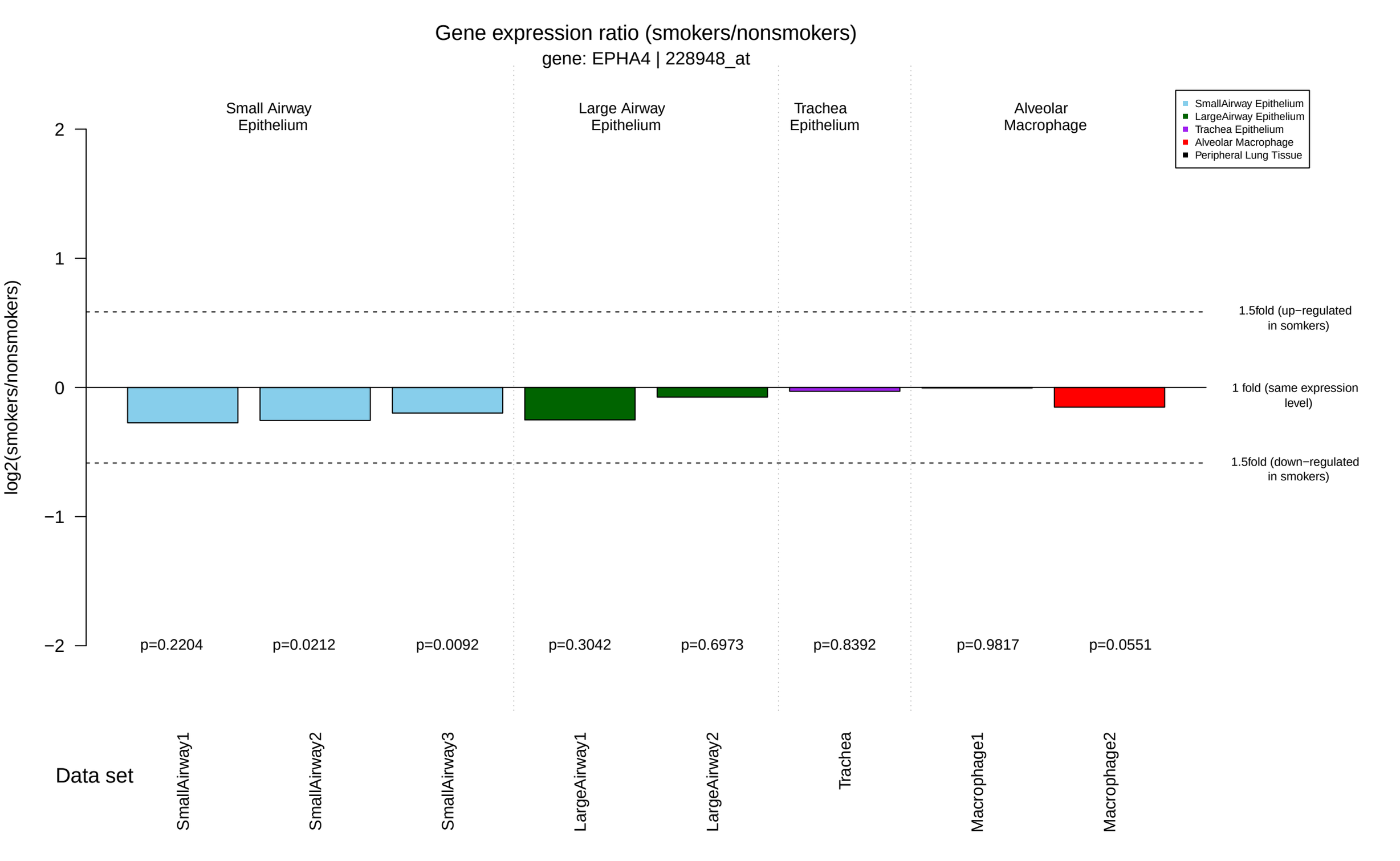
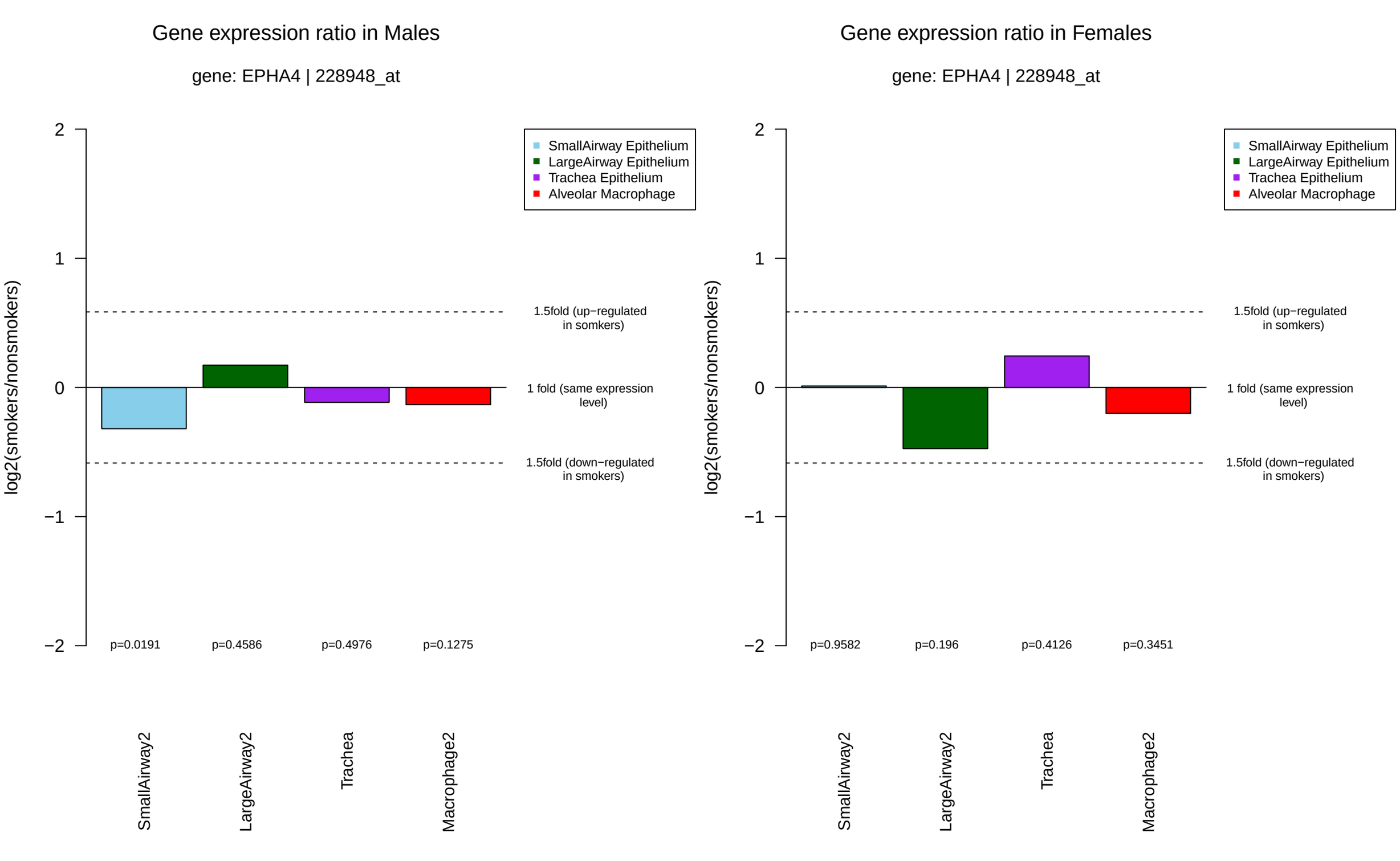
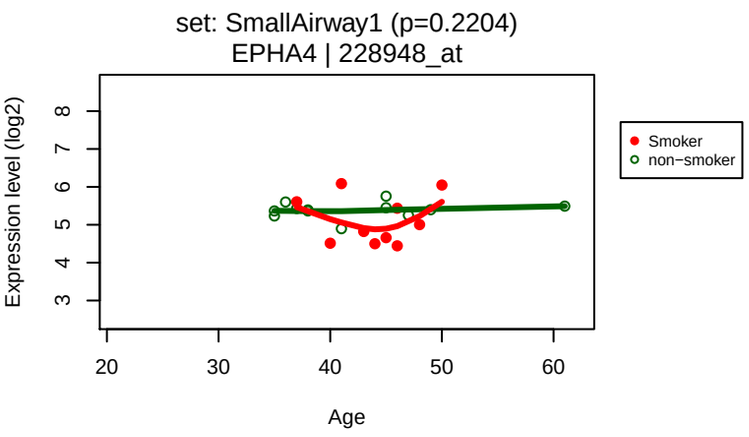
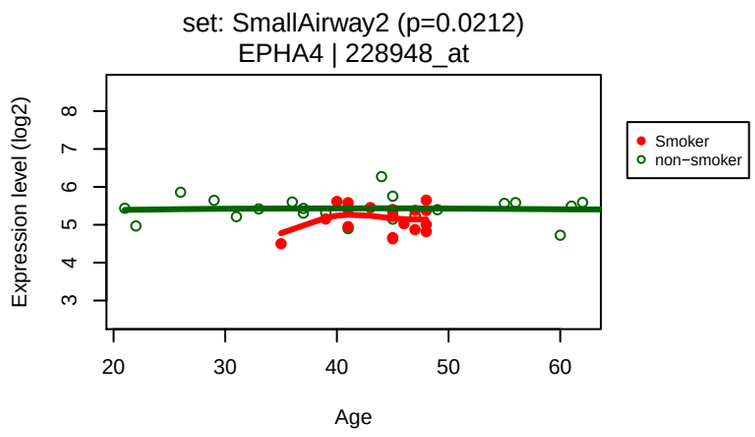
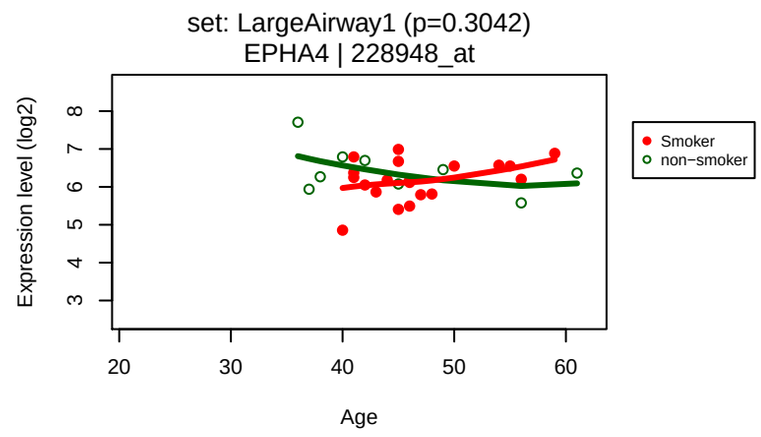

Gene expression of smokers and nonsmokers
gene: EPHA4 | 228948_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2204
p=0.0212
p=0.0092
p=0.3042
p=0.6973
p=0.8392
p=0.9817
p=0.0551
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

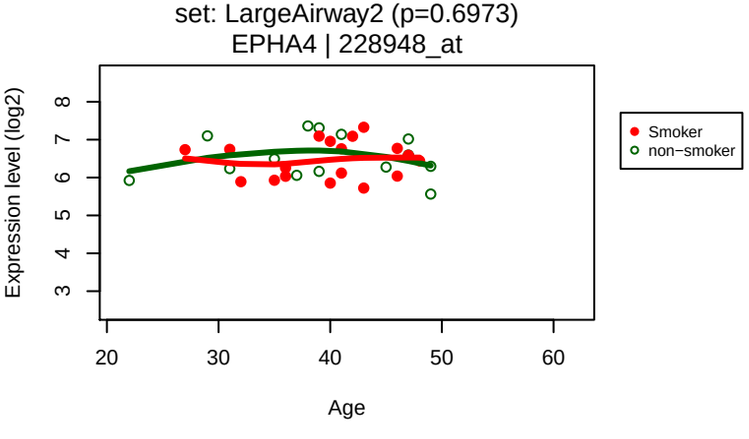

Gene expression ratio (smokers/nonsmokers)
gene: EPHA4 | 228948_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2204
p=0.0212
p=0.0092
p=0.3042
p=0.6973
p=0.8392
p=0.9817
p=0.0551
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

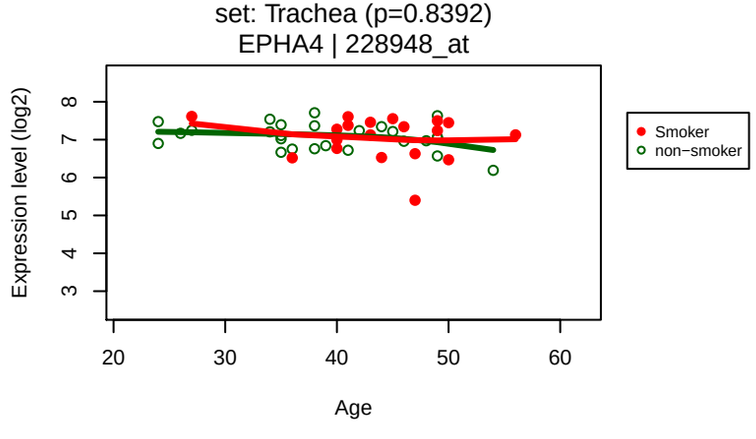

Gene expression ratio in Males
gene: EPHA4 | 228948_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0191
p=0.4586
p=0.4976
p=0.1275
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EPHA4 | 228948_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9582
p=0.196
p=0.4126
p=0.3451
SmallAirway2
LargeAirway2
Trachea
Macrophage2

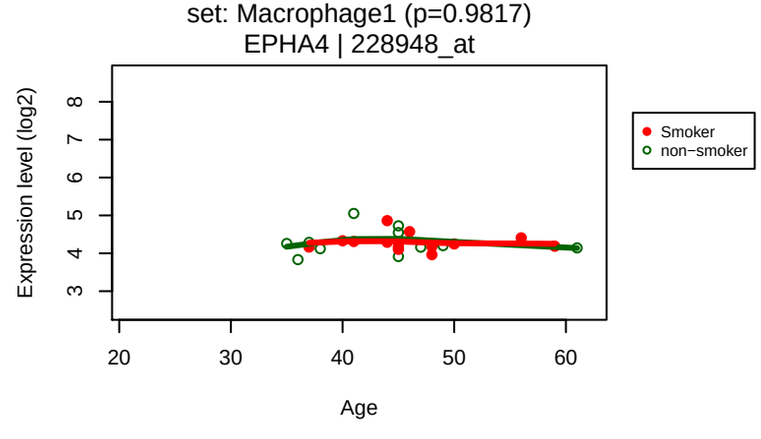

set: SmallAirway1 (p=0.2204)
EPHA4 | 228948_at
Expression level (log2)
Age
Smoker
non−smoker

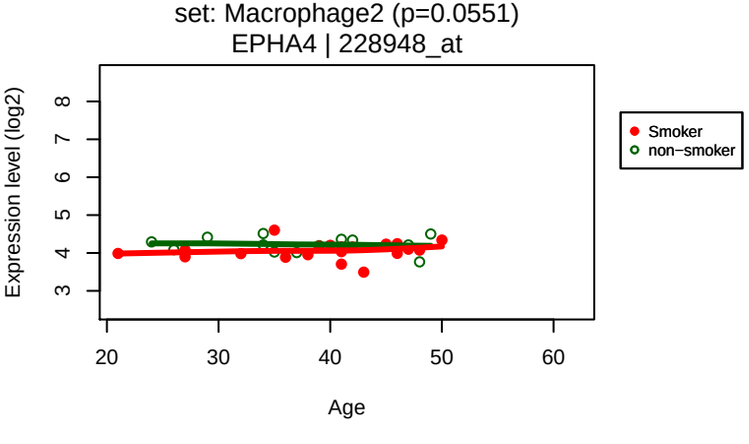

set: SmallAirway2 (p=0.0212)
EPHA4 | 228948_at
Expression level (log2)
Age
Smoker
non−smoker

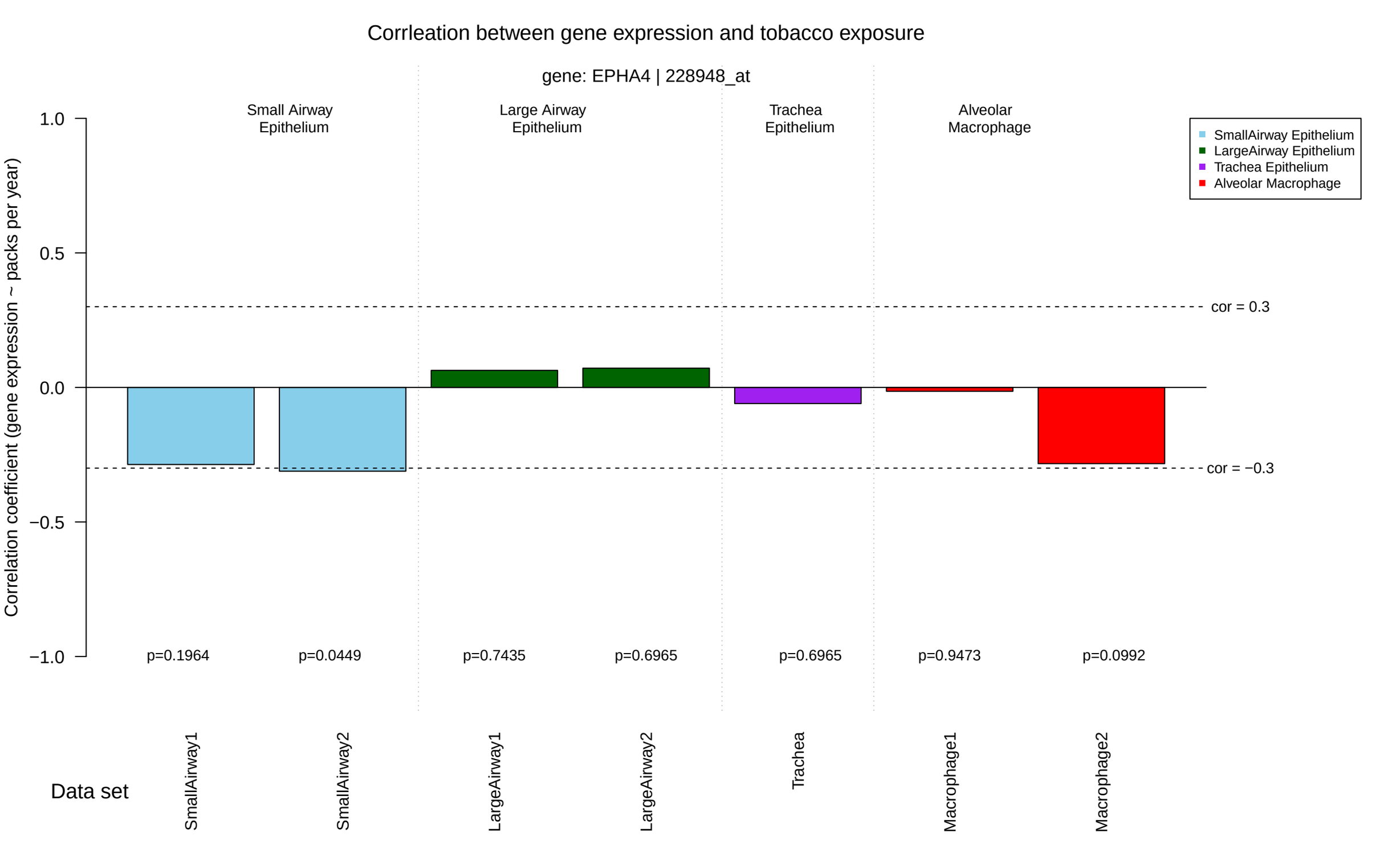

set: LargeAirway1 (p=0.3042)
EPHA4 | 228948_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6973)
EPHA4 | 228948_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8392)
EPHA4 | 228948_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9817)
EPHA4 | 228948_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0551)
EPHA4 | 228948_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EPHA4 | 228948_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1964
p=0.0449
p=0.7435
p=0.6965
p=0.6965
p=0.9473
p=0.0992
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2